Inner Product Spaces

Chapter 1c: Pathak's Introduction to Nonlinear Analysis

Based on H.K. Pathak (2018)

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Outline

Inner Product

Definition (Inner Product)

Let X be a complex linear space. A function $\langle \cdot, \cdot \rangle : X \times X \to \mathbb{C}$ is called an *inner product* if for all $x, y, z \in X$ and $\lambda \in \mathbb{C}$:

IP1 $\langle x, x \rangle \geq 0$ (positivity)

IP2 $\langle x, x \rangle = 0 \iff x = 0$ (definiteness)

IP3 $\langle x, y \rangle = \overline{\langle y, x \rangle}$ (conjugate symmetry)

IP4 $\langle \lambda x + y, z \rangle = \lambda \langle x, z \rangle + \langle y, z \rangle$ (linearity in first argument)

An *inner product space* is a linear space with an inner product.

Consequences of the definition:

- $\langle x, y + z \rangle = \langle x, y \rangle + \langle x, z \rangle$
- $\langle x, \lambda y \rangle = \overline{\lambda} \langle x, y \rangle$

Induced Norm and Cauchy-Schwarz

Theorem (Norm Induced by Inner Product)

For any inner product space, the function $\|\cdot\| : X \to \mathbb{R}_+$ *defined by*

$$\|x\| = \sqrt{\langle x, x\rangle}$$

is a norm on X. *This is called the Hilbertian norm.*

Theorem (Cauchy-Schwarz Inequality)

For all $x, y \in X$,

$$|\langle x, y\rangle| \leq \|x\| \cdot \|y\|$$

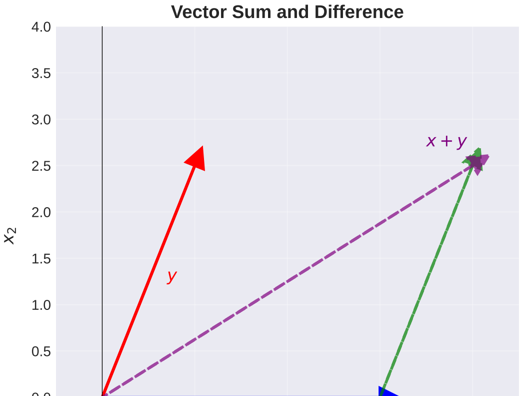
with equality if and only if x *and* y *are linearly dependent.*

Parallelogram Law

Theorem (Parallelogram Law)

For all x, y in an inner product space:

$$2(\|x\|^2 + \|y\|^2) = \|x + y\|^2 + \|x - y\|^2$$



Parallelogram Law

$$\boxed{2(\|x\|^2 + \|y\|^2) = \|x + y\|^2 + \|x - y\|^2}$$

This law characterizes inner product spaces.
A normed space is an inner product space
if and only if the parallelogram law holds.

Orthogonality

Definition (Orthogonal Vectors)

Two vectors x, y in an inner product space are *orthogonal*, denoted $x \perp y$, if

$$\langle x, y \rangle = 0$$

Theorem (Pythagorean Theorem)

If $x \perp y$, then

$$\|x + y\|^2 = \|x\|^2 + \|y\|^2$$

Definition (Orthonormal Set)

A set $\{e_i\}_{i \in I}$ is *orthonormal* if:

- $\langle e_i, e_j \rangle = \delta_{ij}$ (Kronecker delta) for all $i, j \in I$
- $\delta_{ij} = 1$ if $i = j$, and $\delta_{ij} = 0$ if $i \neq j$

Definition of Hilbert Space

Definition (Hilbert Space)

A *Hilbert space* H is a complete inner product space. That is:

- H is an inner product space with inner product $\langle \cdot, \cdot \rangle$
- H is complete with respect to the norm $\|x\| = \sqrt{\langle x, x \rangle}$
- Every Cauchy sequence in H converges to a point in H

In other words, a Hilbert space is a complete Banach space with an inner product structure.

Remark:

- Every Hilbert space is a Banach space
- Not every Banach space is a Hilbert space (e.g., ℓ^p with $p \neq 2$)
- Hilbert spaces have the special feature of being self-dual

Common Hilbert Spaces

Examples of Hilbert Spaces

Example 1: Finite Dimensional

$$\mathbb{R}^n = \{(x_1, \ldots, x_n) : x_i \in \mathbb{R}\}$$

with $\langle x, y\rangle = \sum_{i=1}^n x_i y_i$, $\|x\|_2 = \sqrt{\sum_{i=1}^n x_i^2}$.

Example 2: ℓ^2 Space

$$\ell^2 = \left\{(x_i)_{i=1}^\infty : \sum_{i=1}^\infty |x_i|^2 < \infty\right\}$$

with $\langle x, y\rangle = \sum_{i=1}^\infty x_i \overline{y_i}$ and $\|x\|_2 = \sqrt{\sum_{i=1}^\infty |x_i|^2}$.

Example 3: $L^2[a,b]$ Space

$$L^2[a,b] = \left\{f : [a,b] \to \mathbb{C} : \int_a^b |f(t)|^2 dt < \infty\right\}$$

with $\langle f, g\rangle = \int_a^b f(t)\overline{g(t)}dt$ and $\|f\| = \sqrt{\int_a^b |f(t)|^2 dt}$.

Example 4: Weighted L^2 Space

Numerical Example: The ℓ^2 Space

```python
import numpy as np

# Example: Geometric series sequence
x = np.array([1/2**i for i in range(1, 20)])
y = np.array([1/3**i for i in range(1, 20)])

# Inner product
inner_prod = np.sum(x * y)
print(f"<x, y> = {inner_prod:.6f}")

# Norms
norm_x = np.linalg.norm(x)
norm_y = np.linalg.norm(y)
print(f"||x|| = {norm_x:.6f}")
print(f"||y|| = {norm_y:.6f}")

# Cauchy-Schwarz check
cs_lhs = abs(inner_prod)
cs_rhs = norm_x * norm_y
print(f"|<x,y>| = {cs_lhs:.6f} <= {cs_rhs:.6f}")
print(f"Inequality satisfied: {cs_lhs <= cs_rhs}")

# Pythagorean theorem: Check orthogonality
z = np.array([(-1)**i / 4**i for i in range(1, 20)])
inner_xz = np.sum(x * z)
norm_x_plus_z = np.linalg.norm(x + z)
norm_x_sq_plus_norm_z_sq = norm_x**2 + np.linalg.norm(z)**2
```

Orthogonal Projection onto Subspaces

Theorem (Orthogonal Projection)

Let M be a closed convex subset of a Hilbert space H. For each $x \in H$, there exists a unique point $P_M(x) \in M$ (the projection of x onto M) such that

$$\|x - P_M(x)\| = \min_{y \in M} \|x - y\|$$

Moreover, $P_M(x)$ is characterized by:

$$\langle x - P_M(x), y - P_M(x) \rangle \leq 0 \quad \textit{for all } y \in M$$

Orthogonal Projection - Algebraic Properties

Theorem (Projection Operator Properties)

If M is a closed linear subspace of H, then:

- $P_M : H \to M$ *is a linear operator*
- $P_M^2 = P_M$ *(idempotence):* $P_M(P_M(x)) = P_M(x)$
- $\|P_M(x)\| \leq \|x\|$ *for all* $x \in H$ *(contractivity)*
- $H = M \oplus M^\perp$ *(orthogonal decomposition)*
- $P_{M^\perp} = I - P_M$ *(complementary projection)*

Definition (Orthogonal Complement)

The *orthogonal complement* of M is

$$M^\perp = \{x \in H : \langle x, y \rangle = 0 \text{ for all } y \in M\}$$

Application: Every vector $x \in H$ can be uniquely written as

$$x = P_M(x) + P_{M^\perp}(x)$$

Gram-Schmidt Orthogonalization Process

Theorem (Gram-Schmidt Process)

Let $\{v_1, v_2, v_3, \ldots\}$ *be a linearly independent sequence in an inner product space. Then there exists an orthonormal sequence* $\{e_1, e_2, e_3, \ldots\}$ *such that for each* n*, the span of* $\{e_1, \ldots, e_n\}$ *equals the span of* $\{v_1, \ldots, v_n\}$*.*

Procedure:

Gram-Schmidt Orthogonalization Process

$\boxed{e_1 = \frac{v_1}{\|v_1\|}}$ *Normalize first vector*

$\boxed{u_2 = v_2 - \langle v_2, e_1 \rangle e_1}$ *Orthogonalize second vector*

Complete Orthonormal Systems

Definition (Complete Orthonormal System)

A set $\{e_i\}_{i \in I}$ is a *complete orthonormal system* (CONS) if:

- It is orthonormal: $\langle e_i, e_j \rangle = \delta_{ij}$
- It is complete: if $\langle x, e_i \rangle = 0$ for all $i \in I$, then $x = 0$

Complete Orthonormal System (CONS) in Hilbert Space

An orthonormal set $\{e_i\}_{i \in I}$ is complete if:

$\langle x, e_i \rangle = 0$ for all $i \Rightarrow x = 0$

Parseval's Equality: $\|x\|^2 = \sum_{i \in I} |\langle x, e_i \rangle|^2$

Parseval's Equality

Theorem (Parseval's Equality)

Let $\{e_i\}_{i=1}^{\infty}$ be a complete orthonormal system in a Hilbert space H. Then for all $x \in H$:

$$\|x\|^2 = \sum_{i=1}^{\infty} |\langle x, e_i \rangle|^2$$

Consequence: The Fourier coefficients $c_i = \langle x, e_i \rangle$ form an ℓ^2 sequence.

Example (Trigonometric Basis in $L^2[0, 2\pi)$)

]*Theorthonormalsystem*$\left\{ \frac{1}{\sqrt{2\pi}}, \frac{\cos(nx)}{\sqrt{\pi}}, \frac{\sin(nx)}{\sqrt{\pi}} : n \in \mathbb{N} \right\}$*iscompletein*$\mathrm{L}^2[0, 2\pi]$, yielding the classical Fourier series.

For $f \in L^2[0, 2\pi]$:

$$\int_0^{2\pi} |f(x)|^2 dx = 2\pi a_0^2 + \pi \sum_{n=1}^{\infty} (a_n^2 + b_n^2)$$

Riesz Representation Theorem

Theorem (Riesz Representation Theorem)

Let H be a Hilbert space and $f : H \to \mathbb{C}$ a bounded linear functional. Then there exists a unique $y \in H$ such that

$$f(x) = \langle x, y \rangle \quad \textit{for all } x \in H$$

Moreover, $\|f\| = \|y\|$.

Riesz Representation Theorem

For any bounded linear functional f on a Hilbert space H, there exists a unique $y \in H$ such that

$f(x) = \langle x, y \rangle$ for all $x \in H$

Application: Linear Equations in Hilbert Spaces

Theorem (Lax-Milgram Theorem)

Let $a : H \times H \to \mathbb{C}$ be a bounded bilinear form satisfying:

- *Boundedness:* $|a(x, y)| \leq M\|x\|\|y\|$ *for some* $M > 0$
- *Coercivity:* $|a(x, x)| \geq m\|x\|^2$ *for some* $m > 0$

Then for each bounded linear functional $f : H \to \mathbb{C}$, *there exists a unique* $u \in H$ *such that*

$$a(u, v) = f(v) \quad \text{for all } v \in H$$

Significance:

- Ensures solvability of abstract variational problems
- Applies to weak formulations of PDEs
- Fundamental for numerical methods (finite element methods)

Modes of Convergence

Convergence Concepts in Hilbert Spaces

Strong Convergence	$\|x_n - x\| \to 0$	*Norm convergence*
Weak Convergence	$\langle x_n, y\rangle \to \langle x, y\rangle$ for all $y \in H$	*Weaker notion*

Properties of Weak Convergence

Theorem (Properties of Weakly Convergent Sequences)

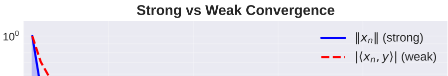
If $x_n \rightharpoonup x$ in H, then:

- *The sequence $\{\|x_n\|\}$ is bounded*
- $\|x\| \leq \liminf_{n\to\infty} \|x_n\|$ *(lower semicontinuity of norm)*
- *The limit x is unique*
- *If $x_n \to x$ strongly, then $x_n \rightharpoonup x$ (strong implies weak)*

Example (Weak Convergence Without Strong Convergence)

In ℓ^2, let e_n be the n-th standard basis vector. Then:

- $e_n \rightharpoonup 0$ (weak convergence)
- $\|e_n - 0\| = 1$ for all n (no strong convergence)



Compactness in Hilbert Spaces

Theorem (Eberlein-Smulian Theorem)

A subset C of a Hilbert space H is weakly compact if and only if it is weakly sequentially compact. That is, every sequence in C has a weakly convergent subsequence.

Theorem (Weak Compactness of Closed Balls)

In an infinite-dimensional Hilbert space:

- *Closed balls are weakly compact but NOT compact (in the strong topology)*
- *Every bounded sequence has a weakly convergent subsequence*
- *This is the basis for many existence results in analysis*

Consequence: Variational problems in Hilbert spaces often have solutions because:

1. Minimizing sequences are bounded
2. Bounded sequences have weakly convergent subsequences
3. Many functionals are weakly lower semicontinuous

Linear Operators in Hilbert Spaces

Definition (Bounded Linear Operator)

A linear operator $A : H_1 \to H_2$ between Hilbert spaces is *bounded* if

$$\|A\| = \sup_{\|x\|=1} \|Ax\| < \infty$$

Definition (Adjoint Operator)

For a bounded linear operator $A : H \to H$, the *adjoint* $A^* : H \to H$ is defined by

$$\langle Ax, y \rangle = \langle x, A^* y \rangle \quad \text{for all } x, y \in H$$

Properties of the adjoint:

- $(A^*)^* = A$ (involution)
- $(A + B)^* = A^* + B^*$
- $(\lambda A)^* = \overline{\lambda} A^*$
- $(AB)^* = B^* A^*$ (reverse order)
- $\|A^*\| = \|A\|$

Special Classes of Operators

Definition (Self-Adjoint Operator)

An operator A is *self-adjoint* if $A = A^*$, i.e.,

$$\langle Ax, y \rangle = \langle x, Ay \rangle \quad \text{for all } x, y \in H$$

Definition (Unitary Operator)

An operator U is *unitary* if $U^*U = UU^* = I$ (isometry and surjective).

Definition (Normal Operator)

An operator A is *normal* if $A^*A = AA^*$.

Definition (Compact Operator)

An operator A is *compact* if it maps bounded sets to relatively compact sets.

Spectral decomposition: Compact self-adjoint operators have countable spectrum with eigenvectors forming a complete orthonormal basis

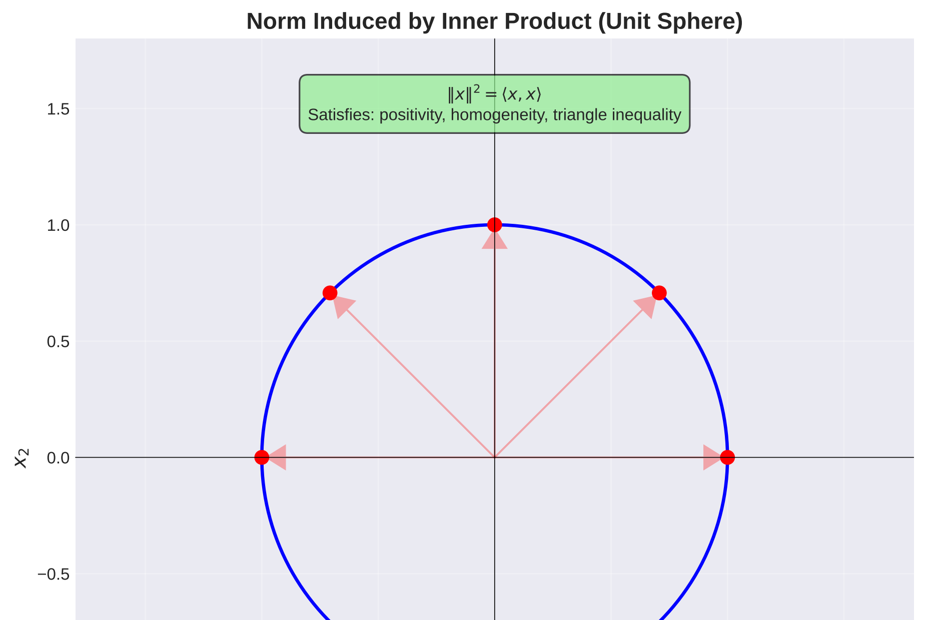
Numerical Example: Operators in ℓ^2

```python
import numpy as np

# Define an orthonormal basis in R^5 approximating l^2
n = 100
e = np.eye(n)  # Standard basis

# Define a self-adjoint operator: diagonal matrix
lamda = np.array([1 - i/n for i in range(n)])
A = np.diag(lamda)

# Test: <Ax, y> = <x, A*y> = <x, Ay>
x = np.ones(n) / np.sqrt(n)
y = np.zeros(n)
y[:10] = 1/np.sqrt(10)

lhs = np.dot(A @ x, y)
rhs = np.dot(x, A @ y)
print(f"<Ax, y> = {lhs:.6f}")
print(f"<x, Ay> = {rhs:.6f}")
print(f"Self-adjoint test passed: {np.isclose(lhs, rhs)}")

# Compute eigenvalues
eigenvalues = np.diag(A)
print(f"\nEigenvalues (first 10): {eigenvalues[:10]}")
print(f"Operator norm ||A|| = {np.max(np.abs(eigenvalues)):.6f}")

# Check spectrum property
```

Applications of Hilbert Space Theory

Quantum Mechanics	*State space as Hilbert space (wave functions)*
Signal Processing	*Fourier analysis, wavelets, time-frequency analysis*
Partial Differential Equations	*Weak solutions, Sobolev spaces*
Optimization	*Convex optimization, projection methods*
Approximation Theory	*Best approximation, least squares*

Induced Norm from Inner Product



Summary: Key Concepts

Inner Product Spaces:

- Definition via four properties (positivity, definiteness, conjugate symmetry, linearity)
- Cauchy-Schwarz inequality: $|\langle x, y\rangle| \leq \|x\|\|y\|$
- Parallelogram law characterizes inner product structure

Hilbert Spaces:

- Complete inner product spaces
- Self-dual (dual space isomorphic to itself via Riesz theorem)
- Examples: $\mathbb{R}^n$, ℓ^2, $L^2[a, b]$

Orthogonal Structures:

- Orthonormal bases exist (Gram-Schmidt process)
- Fourier expansions via complete orthonormal systems
- Parseval's equality relates norms to coefficient sequences

Convergence:

- Strong convergence: $\|x_n - x\| \to 0$
- Weak convergence: $\langle x_n, y\rangle \to \langle x, y\rangle$ for all y
- Weak compactness of closed balls (Eberlein-Smulian)

References and Further Reading

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